Santhosh Narayanan Baburaman

Data Scientist | Machine Learning
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EDUCATION

M.S. Analytics, University of Southern California (GPA: 3.92/4.0) Jan 2025 – Dec 2026
Relevant Coursework: Data Engineering, Database Systems, Enterprise Cloud Analytics, Optimization Methods
B.Tech. Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024
Publication: Co-authored an IEEE conference paper predicting travel time, fare, and speed from 1M+ NYC taxi trips

SKILLS

- **Languages & ML:** Python, SQL, R, PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, NLP, Transformers
- **Statistics:** Mixed-Effects Models, Hypothesis Testing (FDR), ARIMA, Prophet, Anomaly Detection
- **Data & Cloud:** PostgreSQL, BigQuery, Snowflake, Redshift, GCP (Vertex AI, Kubeflow), LLM APIs, Git, Linux

EXPERIENCE

Student Researcher | Prof. Sze-chuan Suen's Lab, USC Viterbi Sep 2026 – Present
- Overhaul the lab's Fitbit sleep pipeline in BigQuery, unifying 7 yearly source tables (2020–2026) into one dataset
- Verify one main sleep per person-night through overlap resolution, cross-source deduplication, and range checks
- Create a second, stitched dataset of continuous nights, rerunning every analysis on both as a robustness check
- Model sleep and cognition links over 1–7 day windows with mixed-effects regression in Python (statsmodels)
- Evaluate predictive signal with XGBoost under GroupKFold cross-validation to prevent person-level leakage

Data Analyst Intern | Philadelphia Department of Aviation Jun 2026 – Aug 2026
- Automated 2 billing data pipelines covering about 30% of airport operating revenue, saving about 60 hours/month
- Established Finance's first structured reporting and data-verification layer, built in Alteryx and Power BI
- Compiled 800K+ billing rows across 4 fiscal years in a Python, Parquet, Alteryx, and Power BI pipeline
- Resolved DAX aggregation and duplicate defects misstating fiscal-year totals in Redshift-based Power BI reports
- Merged both pipelines in Power BI for passenger capture rates, flagging anomalies via Z-scores and Benford's Law
- Documented every Alteryx and Power BI workflow, so Finance runs both pipelines each cycle without support

Founder & President | ISE Analytics Student Association (IASA), USC Feb 2025 – May 2026
- Spearheaded USC's first analytics student organization, hosting 12+ events for 500+ students and 20+ professionals
- Bridged students from diverse cultural backgrounds by fostering an inclusive social and academic community

AI Intern | SmartInternz May 2023 – Jul 2023
- Led a 3-person team building a CNN-LSTM image captioning model in Python with TensorFlow/Keras
- Leveraged a pretrained VGG16 CNN for image features and trained an LSTM decoder on 8,000 Flickr8k images
- Refined 40,000 captions via cleaning, tokenization, and vocabulary building; evaluated outputs with BLEU

PROJECTS

Healthcare Readmissions Pipeline (GCP Vertex AI, Kubeflow, Scikit-learn) May 2026
- Executed data prep in Python for a 30-day readmission model on 8,038 patient records (17% positive class)
- Formulated 3 new features, removed outliers, and applied SMOTE to the training split only to prevent leakage
- Integrated versioned preprocessing into an 8-component Kubeflow pipeline on Vertex AI (team AUPRC 0.86)

Automated Sales Analytics Copilot (Python, PostgreSQL, GitHub Actions, Streamlit) | GitHub Apr 2026
- Forecasted sales KPIs with ARIMA and Prophet over a PostgreSQL star schema ($2.3M in sales, 10K orders)
- Engineered a 4-rule anomaly engine with nightly GitHub Actions runs and Slack and email alerts
- Launched an LLM agent with 6 tools, including text-to-SQL, so non-technical users can query KPIs in plain English

eBay ML Challenge: Product Title NER (Python, PyTorch, NLP) | GitHub Oct 2025
- Developed a BiLSTM-CRF and a fine-tuned DistilBERT for named entity recognition on 30K+ German eBay titles
- Assessed both on brand, model, and color extraction; the BiLSTM-CRF reached 0.78 macro-F1

Consulting Firm Data Warehouse (dbt, Snowflake, SQL, Python) May 2026
- Built a 3-layer dbt ELT pipeline on Snowflake (17 sources, 17 staging models, 3 datamarts) with 3 teammates
- Oversaw the foundation layer, modeling 7 conformed dimensions, including SCD Type 2 history and a date spine
- Wrote 39 automated dbt tests and reconciled design docs against the Snowflake schema, surfacing 13 discrepancies